CP020003 — Artificial Intelligence 2026

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Week 3 — Homework Assignment

The Netflix dataset contains metadata about movies and TV shows available on the Netflix platform, including information such as the director, cast, country, release year, content rating, duration, genres, and descriptions. Although these attributes are primarily intended to help users discover content, they also provide valuable features for machine learning. In this notebook, we will explore the dataset, prepare the data, and build supervised learning models to predict whether a Netflix title is a **Movie** or a **TV Show** based on its metadata.

1. Dataset

netflix-kku-dataset.csv:

```
https://raw.githubusercontent.com/kaopanboonyuen/CP020003_ArtificialIntelligence_2026s1/main/dataset/netflix-kku-dataset.csv
```

Columns: show_id, type, title, director, cast, country, date_added, release_year, rating, duration, listed_in, description

2. Your Task 🎯

Build a **complete, end-to-end supervised learning pipeline** — the same structure we practiced in class — on the Netflix dataset.

Suggested Problem Framing

(you may pick this OR propose your own, clearly justified)

Target variable: type — predict whether a title is a **Movie** or a **TV Show** (binary classification) using metadata *other than* type itself (director, cast, country, release_year, rating, duration, listed_in, description). This is a great beginner-friendly target because:

- It's already a clean binary label (no thresholding needed)
- Several features (like duration format — “90 min” vs “2 Seasons” — and listed_in genres) are strongly predictive, giving you room to feature-engineer
- It mirrors real classification problems: predicting a category from messy, partially-missing metadata

Alternative Target Ideas

(for groups who want a harder challenge)

- Predict rating category grouped into **Kids / Teens / Adults** (multi-class)
- Predict whether a title is_recent (release_year >= 2015) from its metadata (binary)

Required Steps

(mirrors our in-class notebook)

1. **EDA & cleaning** — check missing values (director, cast, country have a lot!), decide how to handle them
2. **Feature engineering** — at least 3 new features (e.g. extract duration_minutes for movies vs duration_seasons for TV shows, n_genres from listed_in, has_director, decade from release_year, etc.)
3. **Encoding** — correctly choose One-Hot vs Label encoding for your categorical features, and justify your choice
4. **Stratified train/test split**
5. **Train at least 5 models** — a mix of Machine Learning models AND at least one Deep Learning model
6. **Evaluate** — with Accuracy, Precision, Recall, F1, ROC-AUC, and a Confusion Matrix
7. **Discuss** — which mistake (False Positive vs False Negative) matters more for your chosen problem, and why
8. **Error analysis** — look at a handful of misclassified rows and hypothesize why the model got them wrong
9. **Conclusion** — which model won, and would you trust it in production? Why or why not?

3. Bonus Points 🌟

- Use an **ML model we did NOT cover in class** (e.g. LightGBM, CatBoost, k-NN, Naive Bayes, Gradient Boosting) — **+bonus**
- Use a **Deep Learning architecture we did NOT cover in class** (e.g. an RNN/LSTM over the description text, a Transformer/embedding-based text classifier, or a wider/deeper network with regularization tricks) — **+bonus**
- Use the description text column with real NLP features (TF-IDF, embeddings, or a Hugging Face model like we did in class with zero-shot classification) — **+bonus**

Good luck, and have fun! 🚀